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The Editor-in-Chief
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Submission of an original research article: "Knowing when to ask a human: fault-aware confidence for fast time series classifiers"

Dear Editor-in-Chief,

The practical problem. Monitoring systems for heart signals, wearables and industrial sensors increasingly let a classifier decide on its own and pass uncertain recordings to a person. This only works if the classifier's confidence drops when a recording is damaged by noise, drift, spikes or dropouts. The fastest accurate time series classifiers, random convolutional kernel methods such as MiniRocket, do not have such a confidence: their existing probability outputs rescale all scores in the same way.

What the paper offers. We propose LOOM, a closed-form readout that scales each prediction by the ridge leverage of the recording, a measure of how far it lies from the training data, and calibrates it on exact leave-one-out quantities without held-out data, refitting or ensembles. We also release an open-source command-line tool that turns labelled recordings into a triage system with a review queue.

The evidence. On 114 UCR datasets with 18 simulated sensor faults, LOOM lowered the median negative log-likelihood from 1.728 to 0.738 compared with the strongest previous readout (PreVal). Under real recorded ECG noise from the MIT-BIH Noise Stress Test Database it lowered it from 3.346 to 0.678. In a pre-registered triage study on 38 sensor datasets, its review queue caught 41 % of the faulty recordings with a 20 % review budget, against 24 % for PreVal. nine of 10 pre-specified hypotheses passed. All numbers are generated by one script from released result files.

Fit with the journal. The work addresses machine learning, data mining and pattern recognition for sensor data, which are within the journal's scope, and offers a computationally light solution for monitoring applications where data are scarce and decisions are shared with a human reviewer.

Main limitations, stated in the paper. On clean data LOOM is not better than PreVal and its calibration error is higher; the tree-based QUANT classifier has a lower likelihood loss under the strongest faults; leverage does not detect changes that keep the features in the training range; apart from the real ECG noise, the faults are simulated.

Declarations. The manuscript is original, has not been published before and is not under consideration elsewhere. All authors have approved the submission [authors: confirm]. The data are public (UCR archive; PhysioNet, ODC-By 1.0). The code, all machine-generated results and the build scripts will be made publicly available [repository link]. The use of an AI assistant is disclosed in the Methods and Acknowledgements, as the journal requires. The manuscript is prepared for double-anonymous review; author details are on a separate title page.

Yours sincerely,

[Corresponding author name], on behalf of all authors